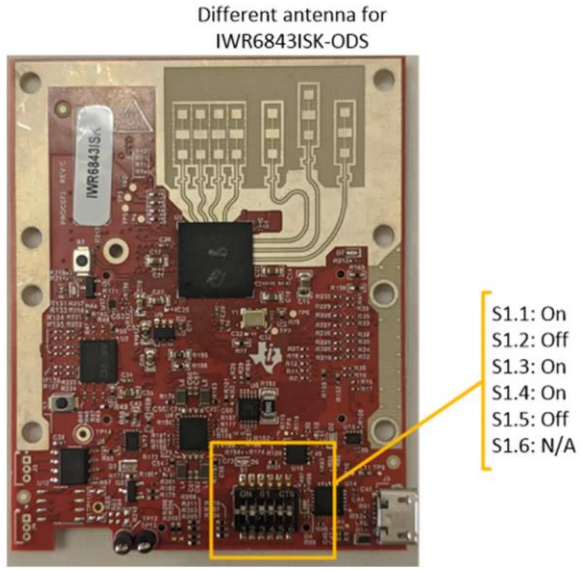
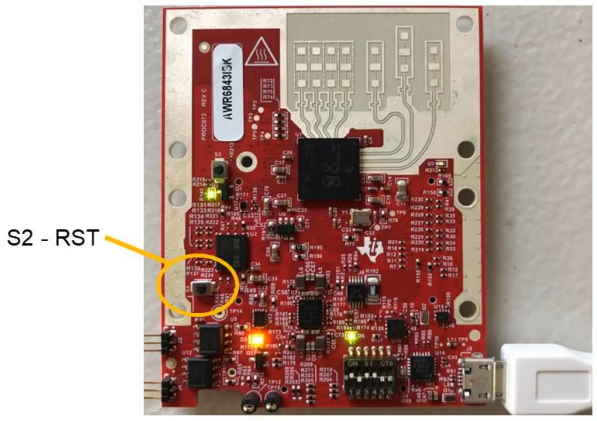
Radar System

Flashing the IWR6843ISK

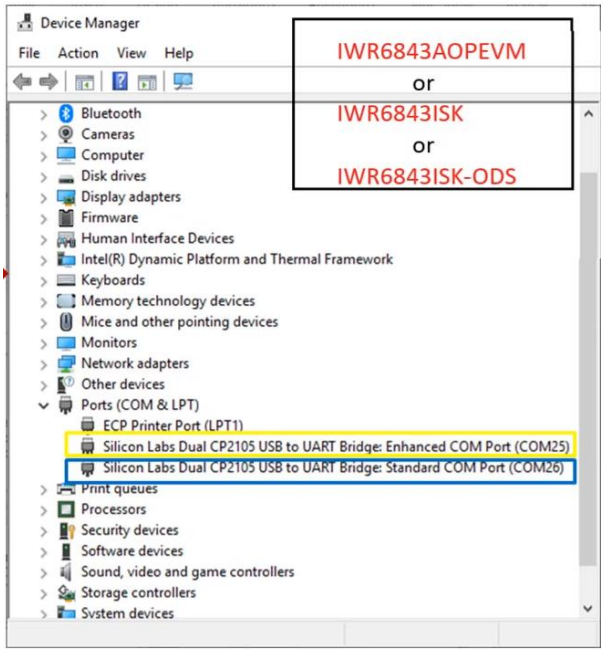
Configure the switches of the antenna module to match the figure below. N/A switches can be turned off.



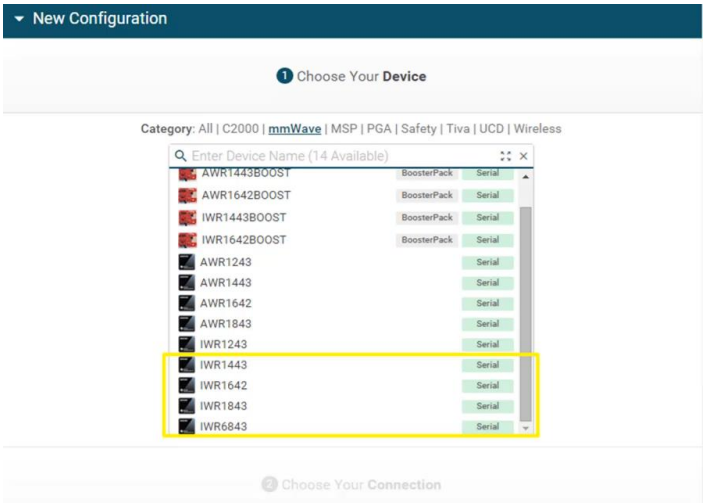
Plug in the micro-USB to the antenna module and your computer, using a computer that uses windows 10 will make this process easier. Then power cycle the device by clicking the RST button, a LED should turn off when pressed and it should now be in flashing mode.



Make sure to download the UniFlash tool. Open device manager and check under Ports (COM & LPT) for the COM number. Take note of the enhanced COM port and open UniFlash.



Select your matching device, IWR6843ISK, then click start.



In settings & utilities, change the COM port to match the enhanced COM port from earlier. In programs under flash image, select the binary file you want to flash then click load image. Make sure no other windows or programs are also connected to the radar when

flashing. UniFlash's console should say Program load completed successfully once it has finished flashing.

For the vital signs we used the binary file vital_signs_68xx.bin from this GitHub repository.

To find the files Texas Instruments provides, download the mmWave software development kit and the radar toolbox. Some file paths of interest from these downloads include:

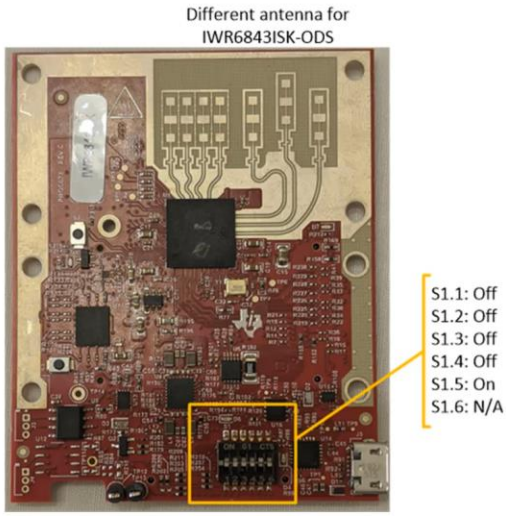
> Other binary programs that show what else the radar is capable of, including people tracking:
> radar_toolbox_3_20_00_04.zip\radar_toolbox_3_20_00_04\source\ti\examples\Industrial_and_Personal_Electronics

> A demo binary file you can run on TI Demo Visualizer to test the radar (in functioning mode after flashing the radar) following this useful video:
> ti\mmwave_sdk_03_06_02_00-LTS\packages\ti\demo\xwr68xx\mmw

Functioning Mode (with MMWAVEICBOOST)

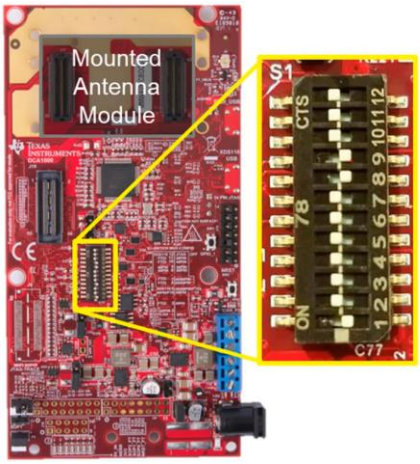
The following instructions are for functional mode for the antenna module with the booster board only. Functional mode for the antenna by itself requires separate instructions otherwise.

Configure the switches of the antenna module to match the figure below. N/A switches can be turned off.



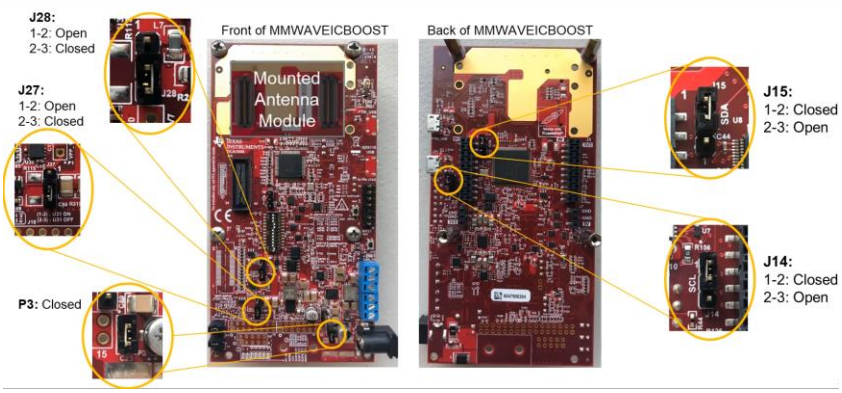
Attach the MMWAVEICBOOST to the antenna module through the 60 pin header to match the figure below.

Configure the switches of the MMWAVEICBOOST to match the figure below. N/A switches can be turned off.

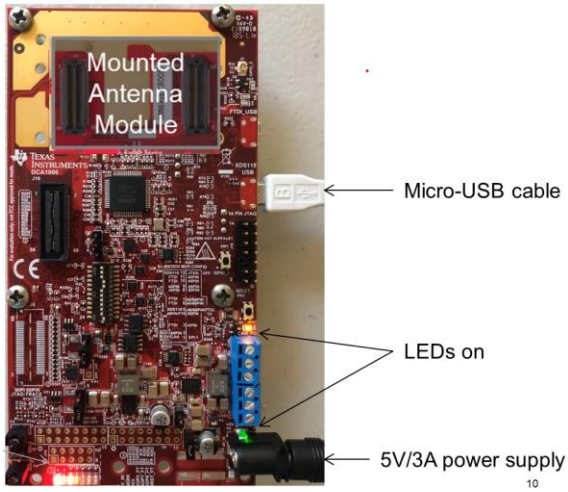


Switch ID	Position
S1.12	ON
S1.11	ON
S1.10	ON
S1.9	OFF
S1.8	OFF
S1.7	ON
S1.6	ON
S1.5	ON
S1.4	ON
S1.3	ON
S1.2	ON
S1.1	OFF

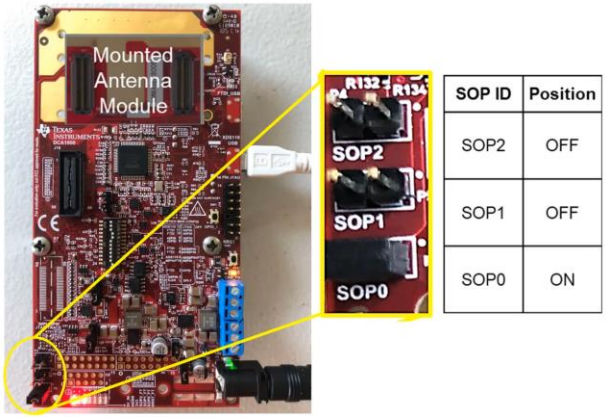
Set the MMWAVEICBOOST jumpers to the default configuration by matching the figure below.



Plug in the micro-USB cable to the computer and the barrel jack to an outlet.

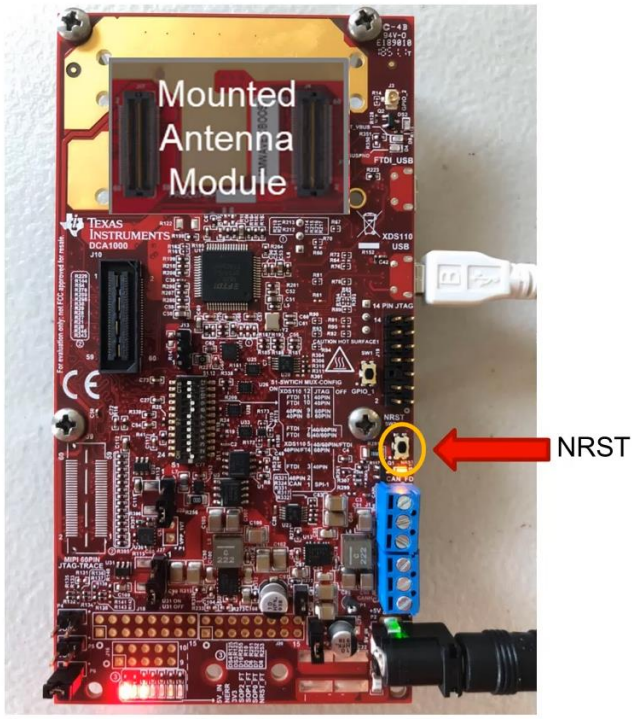


Configure the jumpers below to match for functional mode.



SOP ID	Position
SOP2	OFF
SOP1	OFF
SOP0	ON

Power cycle the device after these changes by pressing the NRST button, while pressed some LEDs will turn off.



Visualizing Radar Data

If you are using the radar for vitals, clone a copy of this Github repository.

If you are using a raspberry pi or a device with multiple projects, make sure to download any needed python packages in a virtual environment by entering the following into the terminal:

```
cd into/the/projects/folder

python3 -m venv env

source env/bin/activate
```

The final source command starts it while closing the terminal window will end the virtual environment. The virtual environment should be used any time python packages are being downloaded or used, or things may break.

Enter the following into terminal to download the needed packages using this format:

Pip install <package name> the following

pyserial

numpy

pyqtgraph

PySide6==6.8.0.2

scikit-learn

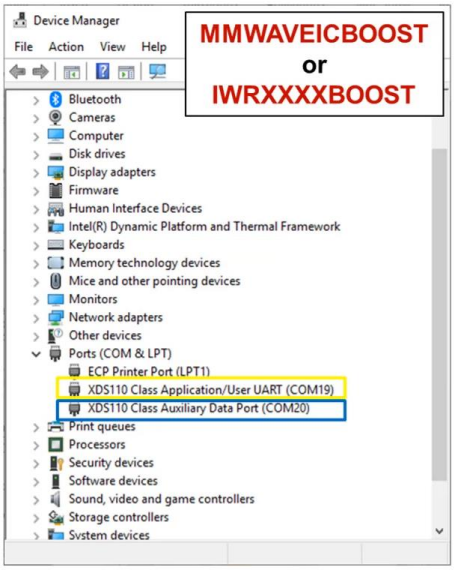
pandas

To run the project, you may need to update the path file in Vitals.py line 19 or the COM PORTS in the mmVS/com.py in line 15 and 16 to match your device.

The raspberry pi COM ports can be found with the following command in terminal.

ls –l /dev/serial/by-id

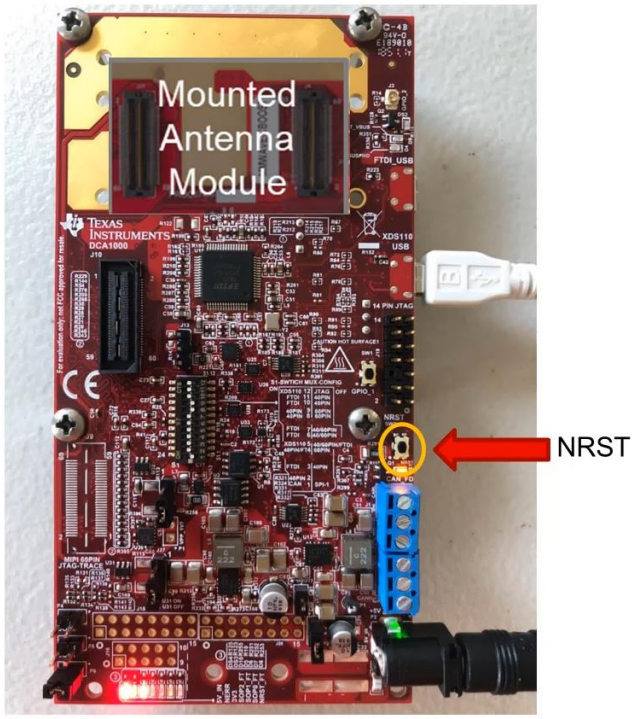
If using a computer with windows, open device manager to note the COM ports, they have likely changed now that the booster board is also attached.



Once everything is downloaded with the virtual environment, press the NRST button than run Vitals.py, it may take a moment to start. On the raspberry pi I used python version 3.11.2 to run the project.

If using a virtual environment, use the following command in the terminal.

python Vitals.py



For other projects, another visualizer application should be used to work properly.

> This visualizer application from the previous downloads can be used with many Texas Instruments binary files:
>
> radar_toolbox_3_20_00_04\radar_toolbox_3_20_00_04\tools\visualizers\Applications_Visualizer\Industrial_Visualizer

Continuing this Project

WHERE TO GO/POSSIBLE IMPROVEMENTS

The current heart and breathing rate do not seem entirely accurate and often do not match what the graphs show, notably when there is nobody in view of the radar. We could make calculations from the graph/raw data to make these calculations ourselves.

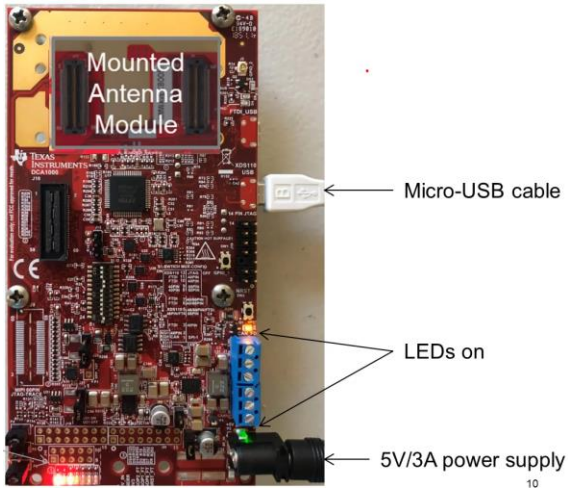
The radar is currently set up on a raspberry pi 3 model B, so it is slow and freezes sometimes. Upgrading the raspberry pi should help with this.

The radar is also capable of tracking people, so this may be an additional feature that could be added.

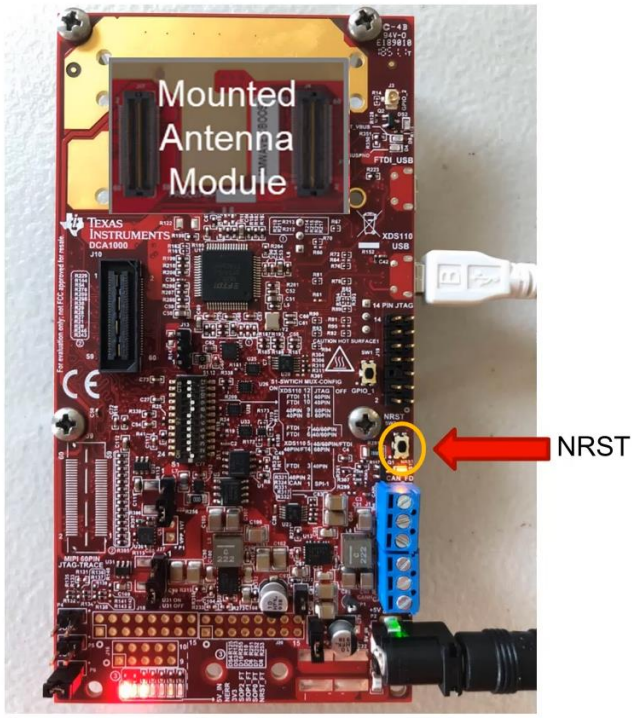
STARTING RADAR VITALS FROM WHERE I SETTED EVERYTHING UP ALREADY

Attach the MMWAVEICBOOST to the antenna module through the 60 pin header to match the figure below.

Plug in the micro-USB cable to the computer and the barrel jack to an outlet.



Power cycle the device by pressing the NRST button, while pressed some LEDs will turn off. This will need to be pressed every time before being used, the graphs will not update otherwise.



Open the terminal on the raspberry pi and enter the following commands:

cd Downloads/otherBranches/Radar-Vitals/

source env/bin/activate

python Vitals.py

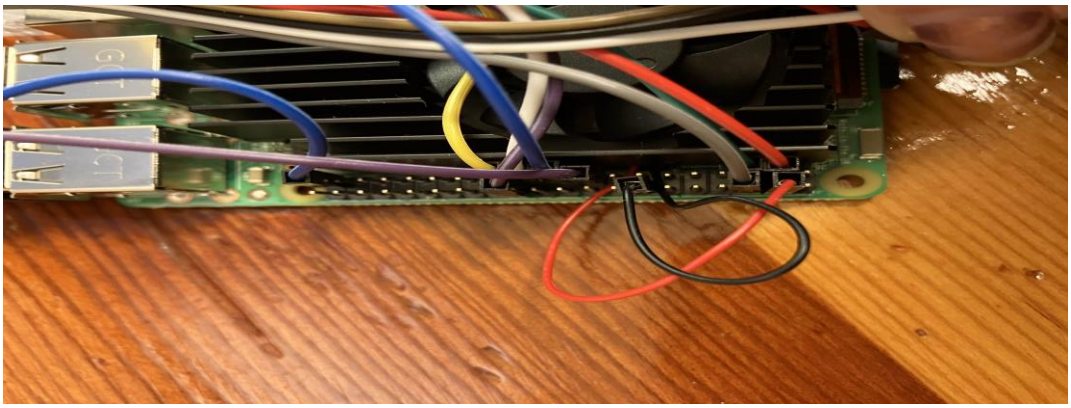
The project will take a moment to start running and will have a delay depending on the device it is running on. It can be especially slow or freeze occasionally when on the raspberry pi, just close or cancel (Ctrl + C) the program, press the NRST button, and rerun it (make sure the virtual environment is still open).

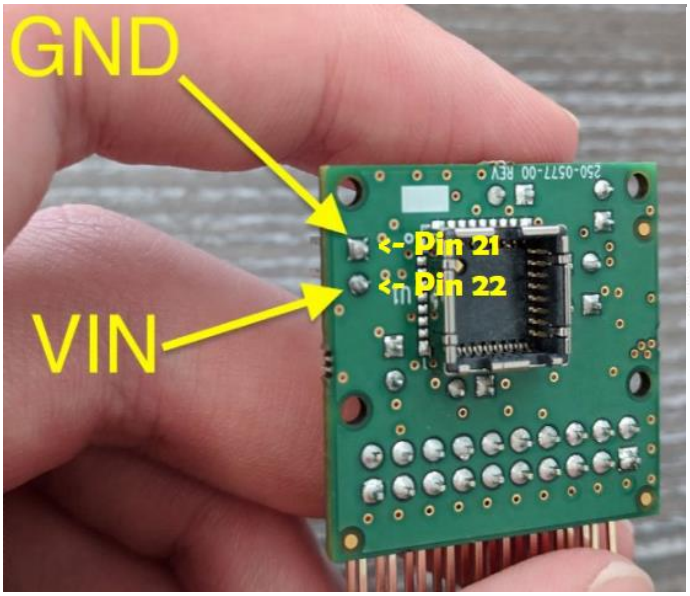
The only lines of code that may need to be changed to start working are the COM ports and file path for the configuration file, as mentioned in the Visualizing data section. This should only be needed if the project is being set up on a new device.

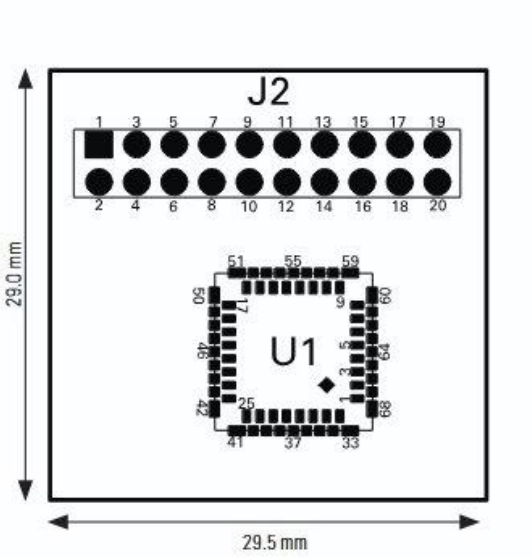
For more accurate data, place the radar vertically facing the chest. It is recommended to stay still for at least 20 or 30 seconds. This video has some related information to this.

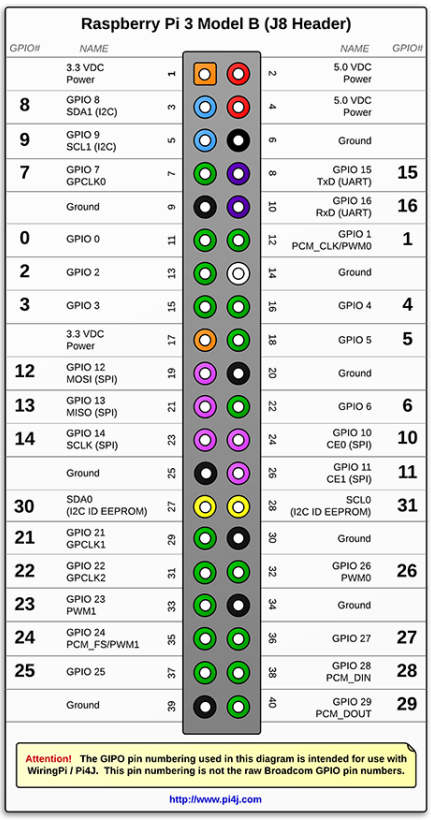
Thermal Camera

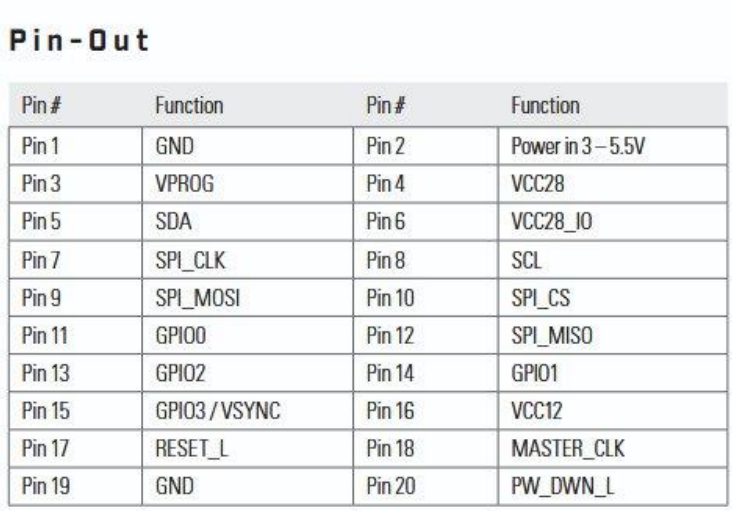
Setting up Flir Radiometric Lepton Dev V2 on a Raspberry Pi 3 model B











Pin-Out

Pin #	Function	Pin #	Function
Pin 1	GND	Pin 2	Power in 3 – 5.5V
Pin 3	VPROG	Pin 4	VCC28
Pin 5	SDA	Pin 6	VCC28_IO
Pin 7	SPI_CLK	Pin 8	SCL
Pin 9	SPI_MOSI	Pin 10	SPI_CS
Pin 11	GPIO0	Pin 12	SPI_MISO
Pin 13	GPIO2	Pin 14	GPIO1
Pin 15	GPIO3 / VSYNC	Pin 16	VCC12
Pin 17	RESET_L	Pin 18	MASTER_CLK
Pin 19	GND	Pin 20	PW_DWN_L

The raspberry pi 3 model B pins should be connected to the cooling fan and FLIR Lepton V2 as listed below; you can use the figures above to help.

RPi J8 pin	→ proper name	→ Lepton J2 pin	→ proper name
Pin 3	→ GPIO 8 SDA1 (I2C)	→ Pin 5	→ SDA
Pin 5	→ GPIO 9 SCL1 (I2C)	→ Pin 8	→ SCL
Pin 6	→ Ground	→ Pin 1	→ Ground
Pin 17	→ 3.3 VDC Power	→ Pin 22	→ VIN
Pin 19	→ GPIO 12 MOSI (SPI)	→ Pin 9	→ SPI MOSI
Pin 21	→ GPIO 13 MISO (SPI)	→ Pin 12	→ SPI_MISO
Pin 23	→ GPIO 14 SCLK (SPI)	→ Pin 7	→ SPI_CLK
Pin 24	→ GPIO 10 CE1 (SPI)	→ Pin 10	→ SPI_CS

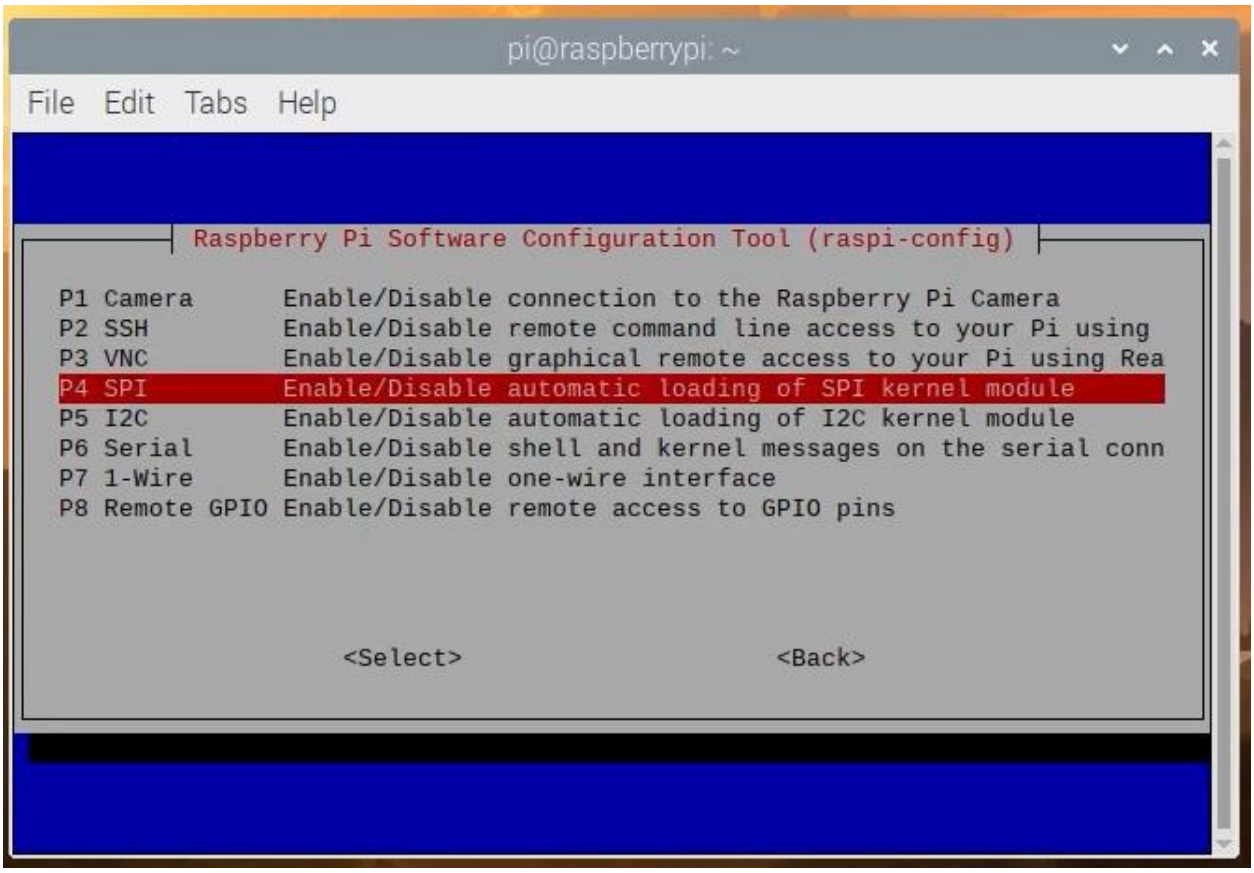
Pin 39	→ GROUND	→ Pin 21	→ GROUND

RPi J8 pin	→ proper name	→ Cooling Fan
PIN 4	→ 5.0 VCD POWER	→ RED
PIN 14	→ GROUND	→ BLACK

Set up the Raspberry Pi as normal with the recommended raspberry pi OS. Then enter the following into the terminal:

sudo raspi-config

Select interface options to configure connections to peripherals and enable I4 SPI and I5 I2C. Once you have selected yes to enable both of them, go back to the terminal screen by selecting finish.



Code Set up

Clone this GitHub repository on to the Raspberry Pi. Follow the directions below to make a virtual environment and install everything from the terminal.

sudo apt-get install qt4-dev-tools

If this doesn't work, try the following commands:

sudo apt update
sudo apt install qtbase5-dev

Cd into the project's folder

python3 -m venv env

source env/bin/activate

The source command starts the virtual environment, make sure to only use or install python packages with the virtual environment or things can break.

pip install ultralytics

The following command should be used in the /LeptonModule/software/raspberrypi_video when you update any file besides yoloDetect.py to see the changes.

qmake && make

The following command shows more options for what the camera can show you.

./raspberrypi_video –h

The following command starts the program, the argument –cm 2 is used because the yolo model was trained on grayscale images.

./raspberrypi_video –cm 2

Click the Perform FFC button to both correct the colors and run the yoloDetect.py file to run the yolo model to detect humans on the camera's current visual. To adjust how the results of the yolo model is shown, adjust this file. The results of what's found will be printed to terminal and a picture with the bounding box will come up with the confidence

score. While running the model the visual for what the camera will freeze, but the thermal camera is still sending updates.

If a black box with red square in top left corner appears when running code, it means there is a problem with the camera, wiring, or that it is already being used (possibly in another window).

In the LeptonThread.cpp file, the temperature can be found with the valueFrameBuffer variable. There is commented out code on line 270 to convert and print this.

Continuing this Project

WHERE TO GO/POSSIBLE IMPROVEMENTS

The project currently just says where it detects humans with a bounding box from the YOLO model. We could find and report the temperatures of the humans in the bounding boxes. We could also set this to run on its own maybe every 30 seconds, this way it could work automatically without a person controlling it.

The current YOLOv8n model was trained on a lot of publicly available data that did not entirely match what the camera would see or have background pictures. While the model can detect people, it can also give more false positives than we would like. A better YOLO version or trained model could be used to fix this. Photos taken to improve the model have been left on the 5 TB hard drive for this.

The Perform FCC button currently color corrects the visual and runs the YOLO model right now. These functions could be separated to be able to do both of these separately from each other.

STARTING THERMAL CAMERA FROM WHERE I SETTED EVERYTHING UP ALREADY

Enter the following commands into the raspberry pi's terminal

cd Downloads/thermal

source venv/bin/activate

cd LeptonModule/software/raspberrypi_video

```
qmake && make

./raspberrypi_video -cm 2
```

Press Perform FCC button to use the YOLOv8n model on the current camera visual. The results of what's found will be printed to terminal and a picture with the bounding box will come up with the confidence score. It may take a moment to show the results.